SmartScalpel

How our tissue classifier works

Reading breast tissue from surgical smoke: the data, the model, how we trained it, and what it found.

Data: Vaysse et al., Scientific Reports 2020 (Maastricht UMC+) · Model: pca-lda-v2 from the SmartScalpel repo

Where the data comes from

689
smoke spectra

72
patients

8,000
m/z channels per spectrum (200–1000)

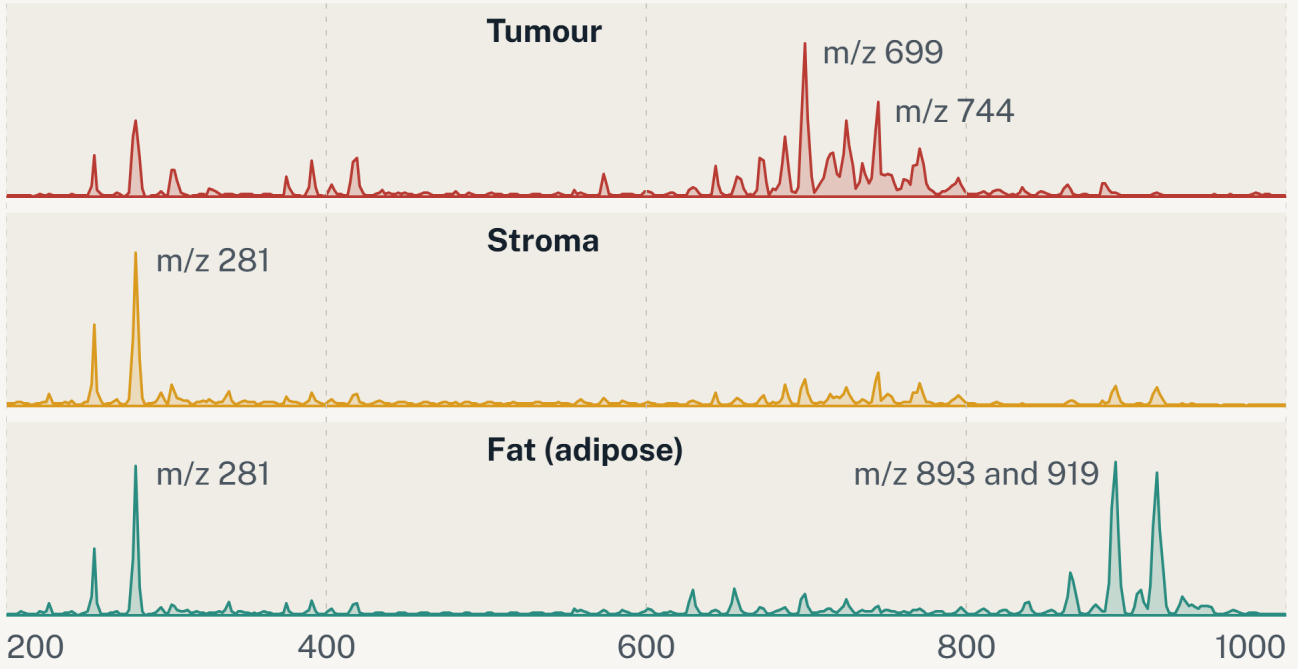
224 · 209 · 256
tumour · stroma · fat spectra

1. **Tissue removed**
   Breast-cancer surgery at Maastricht UMC+, 2017–2019.

2. **Spots chosen**
   A pathologist picks tumour, and healthy tissue at least 2 cm away.

3. **Burned and measured**
   An electrosurgical blade burns each spot; the smoke goes into a REIMS mass spectrometer.

4. **Labelled by pathology**
   The spot is stained (H&E) and labelled by a breast pathologist who never saw the spectrum.



Mean spectrum per tissue, 160 held-out spectra · x-axis: m/z

Source: Vaysse et al., Scientific Reports 2020 (CC BY 4.0) · Chart: our held-out test spectra (spectra_input.csv)

How the model works

Five linear steps turn one smoke spectrum into a tissue call.

Input

8,000 values

Ion intensity at m/z 200–1000, normalised to total ion count

Standardise

8,000 → 8,000

Each channel set to mean 0, SD 1, so small peaks still count

PCA

8,000 → 100

100 lipid-pattern features that keep 89% of the variation

Shrinkage LDA

100 → 3 scores

Picks the tissue cloud that fits best; shrinkage keeps it stable on small data

Confidence check

3 → 1 call

Softmax scores; a top score under 0.77 means ‘uncertain’

≈824,000

learned numbers, which fold into one 3 × 8,000 linear map

1.2 ms

per spectrum in Python; about 0.1 ms folded, in the browser

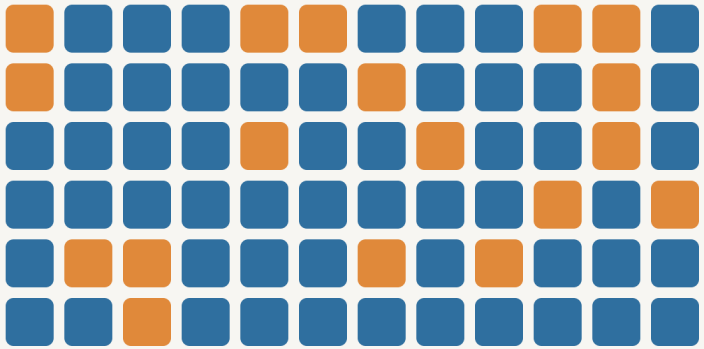
689 spectra

too few for deep learning; PCA-LDA is what the original REIMS studies used

Model: pca-lda-v2 (SmartScalpel repo), scikit-learn 1.9.1 · PCA size chosen by cross-validation from 30 to 150

How we trained and tested it

Split by patient, not by spectrum

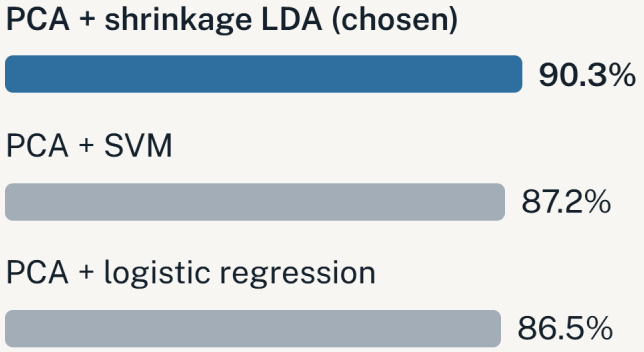


54 training patients · 529 spectra

18 test patients · 160 spectra

All of a patient's spectra stay on one side, so the test measures patients the model has never seen.

Model picked on training patients only



Balanced accuracy in patient-grouped 5-fold cross-validation (bars start at 0%). PCA size tuned from 30 to 150 components; 100 won.

Tested once on 18 new patients

91.25%

accuracy on the held-out patients (146 of 160 spectra)

91.1% ± 2.0

nested cross-validation over all 72 patients (folds 87–94%)

33.8%

with shuffled labels: chance level, so no data leakage

Patient split: GroupShuffleSplit, seed 42 · Threshold and calibration also set on training patients only · reports/evaluation.json

How accurate it is

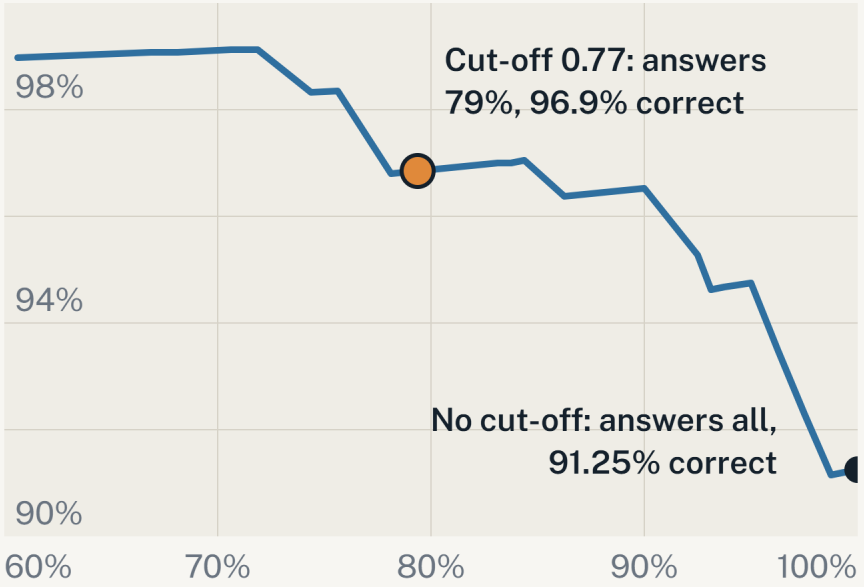
Held-out results, 18 new patients

Rows: true tissue · Columns: the model's call

	Tumour	Stroma	Fat
Tumour	61	2	2
Stroma	2	44	1
Fat	0	7	41

Correct calls in blue: 146 of 160 (91.25%). Recall: tumour 93.8%, stroma 93.6%, fat 85.4%. Biggest error: fat called stroma.

Accuracy vs. how many it answers



Threshold sweep on the 160 held-out spectra. Across: share the model answers. Up: accuracy on those it answers.

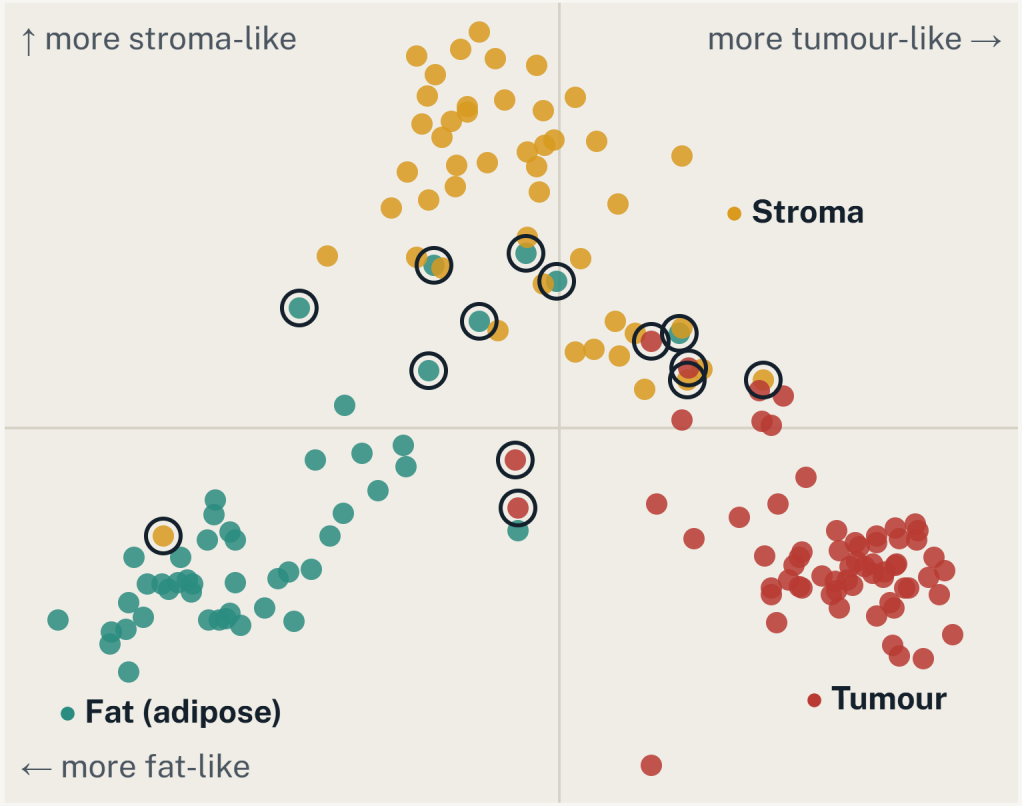
No silent misses

All 4 missed tumour spectra scored below 0.77 (0.48–0.60), so the model flags them 'uncertain' instead of calling them healthy.

Of the 127 spectra it does answer, it misses no tumour and wrongly calls nothing tumour.

Held-out test: 160 spectra from 18 patients never used in training · The 0.77 cut-off was chosen on training patients only

What the model learned



One dot per held-out spectrum · ring = misclassified

Each tissue has its own chemical fingerprint

- **Tumour:** strongest ions at m/z 600–800 (peaks 699, 744, 725), the phospholipid range
- **Stroma:** dominated by m/z 281 and 255, consistent with oleic and palmitic acid
- **Fat:** high-mass ions at m/z 865–920 (893, 919), plus m/z 281

Limits we report

- Fat is the weakest class: 7 of 48 held-out fat spectra were called stroma.
- Ex vivo tissue, one hospital, 72 patients: still needs live-surgery and multi-centre testing.
- Tumour subtype isn't labelled, so no claims per breast-cancer type.

Dots: each held-out spectrum's three classifier scores, projected to 2D · Peaks: held-out mean spectra (1 m/z bins)